

Date: 07 July 2026	Team ID: SB-7429
Project Name: Smart Lender; Applicant Credibility Prediction for Loan Approval	Maximum Marks: 2 Marks

DATA FLOW DIAGRAM TEMPLATE

Data Flow Diagram (DFD) Legend

Understanding the elements of data movement inside the Smart Lender system.

Symbol Shape	DFD Element Name	Description / Purpose in System
Oval / Rounded Shape	External Entity	The Loan Applicant or Credit Officer who inputs profile details and receives eligibility reports.
Rectangle with Header	Process	Logic processing stages: 1.0 Impute & Preprocess, 2.0 Feature scaling, 3.0 XGBoost prediction.
Rectangle (solid borders)	Data Store	Static files storing training dataset (train.csv) and trained weights (loan_model.pkl, scaler.pkl).
Labeled Arrow	Data Flow	Represents data passing between entities, processes, and stores (e.g. features vector, prediction decision).

Level 1 DFD Detailed Steps

The detailed flow comprises the following processing steps:

Process ID	Source Element	Process Description	Data Store / Destination
1.0	home.html form	Sanitizes and checks form numeric limits (e.g. positive values for income and loan).	Passes features dictionary to Process 2.0.
2.0	Process 1.0 dict	Encodes categorical features and applies StandardScaler transformations.	Passes scaled numpy array to Process 3.0.
3.0	Process 2.0 array	Loads loan_model.pkl (XGBoost) and runs classification inference.	Calculates decision and probability vector.
4.0	Process 3.0 outputs	Renders approval/rejection banner and logs details on result.html.	Sends final HTTP response to user browser.